Aim:Basic MANET Implementation with Packet Animation
Simulate a mobile ad-hoc network and visualize packet movement to study dynamic topology formation.
Manet.tcl

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# ======================================================================
# Basic MANET Simulation with Packet Animation
# ======================================================================

set val(chan)       Channel/WirelessChannel
set val(prop)       Propagation/TwoRayGround
set val(netif)      Phy/WirelessPhy
set val(mac)        Mac/802_11
set val(ifq)        Queue/DropTail/PriQueue
set val(ll)         LL
set val(ant)        Antenna/OmniAntenna
set val(ifqlen)     50
set val(nn)         5
set val(rp)         AODV
set val(x)          500
set val(y)          500
set val(stop)       30

# Simulator
set ns [new Simulator]

# Trace Files
set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

# Topography
set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)

# GOD Object
create-god $val(nn)

# Channel
set chan_1_ [new $val(chan)]

# Wireless Configuration
$ns node-config \
    -adhocRouting $val(rp) \
    -llType $val(ll) \
    -macType $val(mac) \
    -ifqType $val(ifq) \
    -ifqLen $val(ifqlen) \
```

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    -antType $val(ant) \
    -propType $val(prop) \
    -phyType $val(netif) \
    -channel $chan_1_ \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON \
    -movementTrace ON

# Create Nodes
for {set i 0} {$i < $val(nn)} {incr i} {
    set node_(i)[ns node]
}

# Initial Positions
$node_(0) set X_ 50
$node_(0) set Y_ 50
$node_(0) set Z_ 0

$node_(1) set X_ 150
$node_(1) set Y_ 150
$node_(1) set Z_ 0

$node_(2) set X_ 250
$node_(2) set Y_ 250
$node_(2) set Z_ 0

$node_(3) set X_ 350
$node_(3) set Y_ 350
$node_(3) set Z_ 0

$node_(4) set X_ 450
$node_(4) set Y_ 450
$node_(4) set Z_ 0

# Display Nodes in NAM
for {set i 0} {$i < $val(nn)} {incr i} {
    node(i) random-motion 0
    $ns initial_node_pos node(i) 20
}

# Mobility
nsat2.0"node_(0) setdest 400 100 15"
nsat4.0"node_(1) setdest 450 250 10"
nsat6.0"node_(2) setdest 100 400 12"

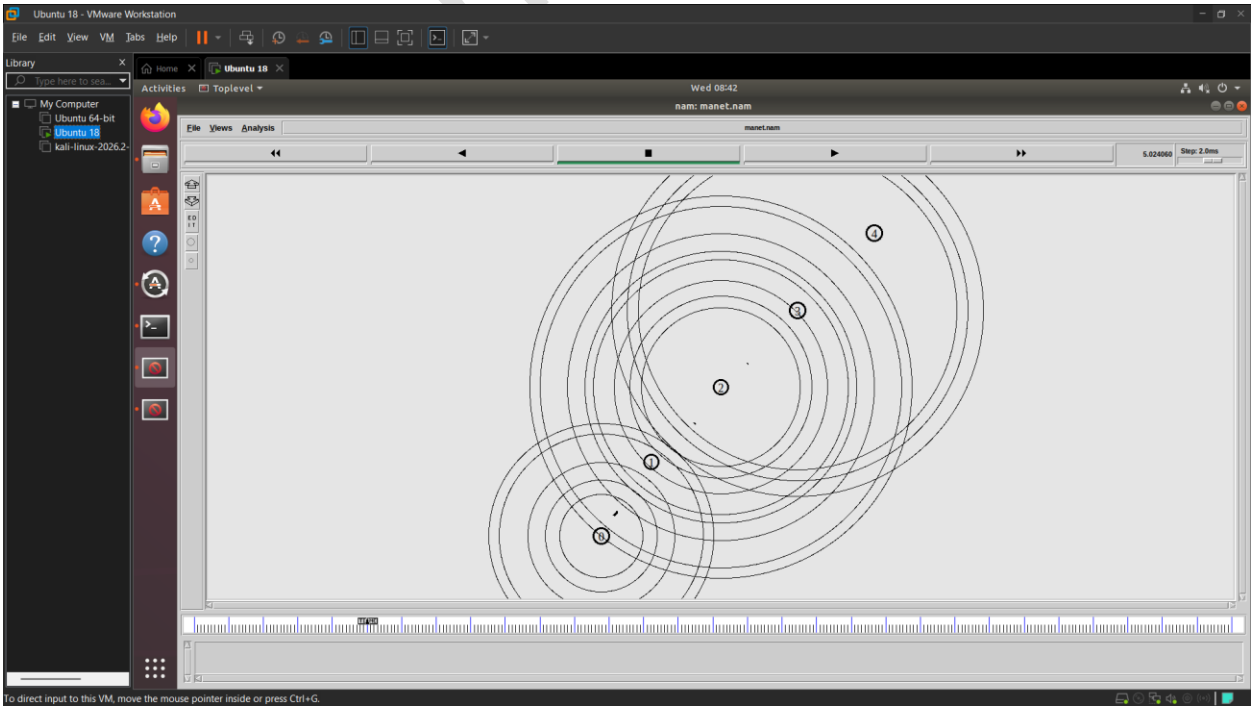
# UDP Agent
set udp0 [new Agent/UDP]
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$ns attach-agent $node_(0) $udp0

set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0
$ns connect $udp0 $null0
# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set rate_ 200k
$cbr0 attach-agent $udp0
nsat5.0"cbr0 start"
nsat25.0"cbr0 stop"
# End NAM
$ns at val(stop)"ns nam-end-wireless $val(stop)"
# Finish Procedure
proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    puts "Simulation Completed"
    exec nam manet.nam &
    exit 0
}
$ns at $val(stop) "finish"
puts "Starting MANET Simulation..."
$ns run
```

Output:



Result

The MANET network was successfully simulated using NS2. Dynamic node movement and packet transmission were observed through NAM animation. The AODV routing protocol successfully established routes between mobile nodes.

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